

Software program

The software flowchart programmed in the microcontroller of the transmitter section is shown in Fig. 6. It is written in Assembly language and compiled using ASM51 software to generate the hex code. The hex program can be burnt into the AT89C51 microcontroller by using any standard programmer available in the market. We have used TopView programmer from Frontline Electronics to program the microcontroller.

The software program is designed to accept the input from the user as well as control the devices. It identifies the key pressed and displays the key code on the LCD module.

In the program, the LCD module is initialised first. As soon as the time-out is set, all the four devices turn on initially, then a particular device turns off at preset time. In this project, the time-out range is 00 to 99 seconds, which can be easily modified to extend the time duration in the delay subroutine

of Assembly code.

Port 0 is configured as output port and interfaced with the RF module through encoder IC1. Port 1 is used for LCD interface and port 2 is used for the input from push-to-on switches.

Circuit operation

When the system is switched on, the startup message "press any key" appears on the LCD screen. When any key is pressed by the user, the LCD displays the message "to set time out press ent!". Pressing 'ent' key displays the following messages on the LCD with a cursor blinking near the first device 'D1_T':

D1_T= D2_T=
D3_T= D4_T=

Use 'up' and 'down' keys to set the time for controlling the devices. The set time for each device on the LCD screen looks like this:

D1_T=10 D2_T=20
D3_T=30 D4_T=40

Now press 'ent' key followed by 'run' key. A device control subroutine

executes and sends the data to the RF module, which transmits the data through ANT antenna. You can set maximum of 99 seconds as the control time for the device. If you set it to 00, a particular device is turned on for infinite time.

Construction

An actual-size, single-side PCB layout of the transmitter for wireless equipment control using microcontroller is shown in Fig. 7 and its component layout in Fig. 8. The actual-size, single-side PCB layout for the receiver circuit is shown in Fig. 9 and its component layout in Fig. 10.

EFY. The source codes of this project have been included in this month's EFY-CD and are also available on www.efymag.com website. ●

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